



ONLINE VEHICLE ACCESS MONITORING

**PHUMPHOL CHANRUNGRICHAY
NUTCHANON THAMAVUTICHAJ**

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2020

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

ONLINE VEHICLE ACCESS MONITORING

**PHUMPHOL CHANRUNGRICHAY
NUTCHANON THAMAVUTICHA**

A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2020

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

ONLINE VEHICLE ACCESS MONITORING

**PHUMPHOL CHANRUNGSRICHAY
NUTCHANON THAMAVUTICHA**

THIS SENIOR PROJECT HAS BEEN APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING
2020

EXAMINING COMMITTEE

.....
(Aj. Surapol Vorapatratorn) **ADVISOR**

.....
(Aj.Dr. Suppakarn Chansareewittaya) **COMMITTEE**

.....
(Aj.Dr. Mahamah Sebakor) **COMMITTEE**

ACKNOWLEDGEMENT

This project will not be successful without the kind support of the advisor, committees and all supportive persons. We would like to thank everyone for their advice and help. We are also grateful to our parents and friends for their continuous encouragement throughout this project. This accomplishment would not have been possible without them. Thank you.

Phumphol Chanrungsrichay
Nutchanon Thamavuthichai

Title	Online Vehicle access monitoring	
Author	Mr. Phumphol Chanrungrichay	
	Mr. Nuchanon Thamavuthichai	
Degree	Bachelor of Engineering (Computer Engineering)	
Supervisory Committee	Aj. Surapol Vorapatratorn	Advisor
	Aj.Dr. Suppakarn Chansareewittaya	Committee
	Aj.Dr. Mahamah Sebakor	Committee

Abstract

This senior project proposes a development of the way to access vehicle for Mae Fah Luang University. The way change to use the RFID technology and get the data of entry and exit. The system have web application for user to request the access card online. Moreover, user can view their access information. Admin can view all of access information. It makes if any problem happens admin can provide the information for help to solve problem. But the main propose of this system is decrease the traffic jam in entry and exit and makes user easily to know the information.

The results showed that the developed application is complete as indicated in the scope. In addition, the survey of 15 users revealed that the application could help both experienced and inexperienced users to understand 3D computer graphics in a good level.

Keyword: RFID

CONTENTS

ACKNOWLEDGEMENT	iii
ABSTRACT	iv
LIST OF TABLES	vi
LIST OF FIGURES	vii
CHAPTER 1 INTRODUCTION	9
1.1 Background and Rational	9
1.2 Objective	10
1.3 Scope	10
1.4 Methodology	10
1.5 Plan	11
1.6 Expected Result	11
1.7 Place and Equipment	12
CHAPTER 2 LITERATURE REVIEW	13
2.1 Related Work	13
2.2 Related Technology	13
CHAPTER 3 ANALYSIS AND DESIGN	15
3.1 Existing System	15
3.2 New System	15
3.3 Activity diagram	16
3.4 Use case diagram	19
3.5 Database	20
3.6 Use Interface Design	21
CHAPTER 4 EXPERTIMENT AND RESULT	28
4.1 Hardware	28
4.2 User Web Application	29
4.3 Admin Web Application	34
CHAPTER 5 CONCLUSION AND DISCUSSION	35
REFERENCES	36

LIST OF TABLES

Table	Page
Table 1.1 Working plan	10

LIST OF FIGURES

Figure	Page
Figure 2.1 RFID	14
Figure 3.1 Block diagram of the system	15
Figure 3.2 Activity diagram of Registration the account of web application	16
Figure 3.3 Activity diagram of requesting the access card	17
Figure 3.4 Activity diagram of check and approval the vehicle information	18
Figure 3.5 Activity diagram of scan access card	19
Figure 3.6 Use case diagram of online vehicle access monitoring of MFU	19
Figure 3.7 Home page	20
Figure 3.8 Registration form	20
Figure 3.9 Pop-up finished registration	21
Figure 3.10 Page after login	21
Figure 3.11 User access information	22
Figure 3.12 User Information	22
Figure 3.13 Request access card form	23
Figure 3.14 Pop-up Request access card form	23
Figure 3.15 Forget password	24
Figure 3.16 Pop-up Forget password	24
Figure 3.17 Create new password	25
Figure 3.18 Pop-up Create new password	25
Figure 3.19 Statistic of all vehicle access	26
Figure 3.20 Contact	26
Figure 3.21 Home Page (User part)	27
Figure 3.22 Contact Page (User part)	27
Figure 3.23 Login Page (User part)	28
Figure 3.24 Register Page (User part)	28
Figure 3.25 Completed register page (User part)	29
Figure 3.26 Home page after login (User part)	29

Figure 3.27 Your statistics page (User part)	29
Figure 3.28 Your information (User part)	30
Figure 3.29 Request access card page (User part)	30
Figure 3.30 Home page (Admin part)	31
Figure 3.31 Request Access Card page (Admin part)	31
Figure 3.32 Database	32
Figure 3.33 Arduino IDE Serial monitor	32

CHAPTER 1

INTRODUCTION

1.1 Background and Rational

Nowadays, vehicles are essential to travel, whether it be a bicycle, a motorcycle, a car, or bus. It is used for both private and public. In the places with a lot of people living, there is an increase in vehicle consumption.

Mae Fah Luang University is one of the universities in Chiang Rai with a large number of students. Statistics from the Division of Register of Mae Fah Luang University showed that there were 13,852 undergraduate, master and doctoral students, excluding the number of professors and university employees. Due to a lot of people, there is also a large vehicle utilization rate. Moreover, Mae Fah Luang University has only one arrival and departure gate. There was a traffic jam during the entrance in the morning, during exit at noon and during exit at time off work despite opening multiple traffic lanes to enter and exit. In addition, at the entrance there is a car card acceptance and the exit still has the car card returned. This has caused many problems with lost cards, risk of accident, and high exposure of the car card making it vulnerable to various infections. Especially at present with the COVID-19. At present, Mae Fah Luang University provides another way for students and employees to get access. It changes from card exchange to vehicle registration. People that already registered the car have to stick the sticker that shows the vehicle number in front of the car. People who have registered this access card will be able to enter and exit the university without having to exchange any card. This way reduces traffic congestion while card exchange. But this sticker cannot provide the time of entry and exit of each vehicle, making it still lacking safety. In addition, the sticker must be registered at the building division. Some people may be inconvenienced to submit documents and therefore still use the card exchange to enter and exit university. As a result, the problem remains the same.

This project aims to develop an innovation to improve the vehicle access management system. This system will make people easy to enter to university by register the car to the system and then getting the access card for the car. The access card made from RFID technology. People have scan access cards with RFID readers and the system will send data to the server. The system will collect the data and can show the time stamp of vehicle access then keep it in the database and then report it.

1.2 Objective

1.2.1 The system can solve traffic jam problems.

1.2.2 The system runs without problem and works well.

1.2.3 The staff can use this system.

1.2.4 The RFID system can work.

1.2.5 The system can show the timestamp of car enter and exit.

1.3 Scope

1.3.1 General user

- 1) Get information of user from register
- 2) Register car information and get RFID card
- 3) Check time while enter and exit University

1.3.2 Administrator

- 1) Receive user information
- 2) Update information and delete old information

1.3.3 Staff

- 1) Scan RFID while car or motorcycle entrance and exit
- 2) Send information to web server for update

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Literature review and requirement gathering

1.4.3 System design

1.4.4 Design web interface

1.4.5 Computer Engineering Pre-Project document preparation

1.4.6 Computer Engineering Pre-Project final presentation

1.4.7 Start coding service

1.4.8 Combine all part of system

1.4.9 System testing

1.4.10 Computer Engineering Project document preparation

1.4.11 Computer Engineering Project final presentation

1.5 Plan

Table 1.1 Working plan

	Pre-project					Project				
Plan/Month	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Proposal preparation and defense										
Literature review and requirement gathering										
System design										
Design web interface										
Pre-project document preparation										
Pre-project final presentation										
Start coding service										
Combine all part of system										
System testing										
Project document preparation										
Project final presentation										

1.6 Expected Result

1.6.1 MFU Online vehicle access monitoring web application.

1.6.2 The system can help people easy to register their information.

1.6.3 The system will help to reduce traffic jams.

1.6.4 The system will keep the data of vehicle access.

1.6.5 The system reduce the chance of COVID-19

1.6.6 The system reduce an accident while get and return card

1.7 Place and Equipment

1.7.1 Place

- Dormitory
- MFU

1.7.2 Equipment

1.7.2.1 Hardware

- RFID
- Personal compute

1.7.2.2 Software

- Visual Studio Code
- Xaamp
- Adobe XD
- MySQL workbench

1.7.3 Budget

-

CHAPTER 2 LITERATURE REVIEW

2.1 Related Work

Table 2.1 Related work

Author	Topic	Tool
Kittiphong et al.	RMUTT Vehicle license plates detection and information system	Image processing
Chaithawhach M (2016)	Automatic License Plate Recognition System	Image processing
Smart campus Management center of CMU	Chiangmai University smart pass	Barcode

To detect the vehicle, this part shows the tools that can be used to detect the vehicle when entering and exiting university.

Kittiphong et al. (2012) and Chaithawhach M (2016) creating the device with sensors work together with the camera to detect the license plate by image processing and bring the data that received from the previous step to keep in the database. The vehicle owner has to register the car to the system before. And the system can show the history of entry and exit by choosing the calendar. But the result of processing the license plate cannot be 100% correct. The image processing cannot work well in the dark. And sometimes that the characters in the license plate are similar, the image processing always types the wrong character. So, this research will not use image processing to detect the vehicle but also use the way to register the vehicle to collect the vehicle information.

2.2 Related Technology

Identify both hardware and software required for the project as well as their references. The details should include the target version and you could explain each item separately as a new subtopic.

2.3.1 RFID radio-frequency identification

RFID is an acronym for “radio-frequency identification” and refers to a technology whereby digital data encoded in RFID tags or smart labels (defined below) are captured by a reader via radio waves. RFID is similar to barcoding in that data from a tag or label are captured by a device that stores the data in a database. RFID, however, has several advantages over systems that use barcode asset tracking software. The most

notable is that RFID tag data can be read outside the line-of-sight, whereas barcodes must be aligned with an optical scanner. If you are considering implementing an RFID solution, take the next step and contact the RFID experts at AB&R® (American Barcode and RFID).

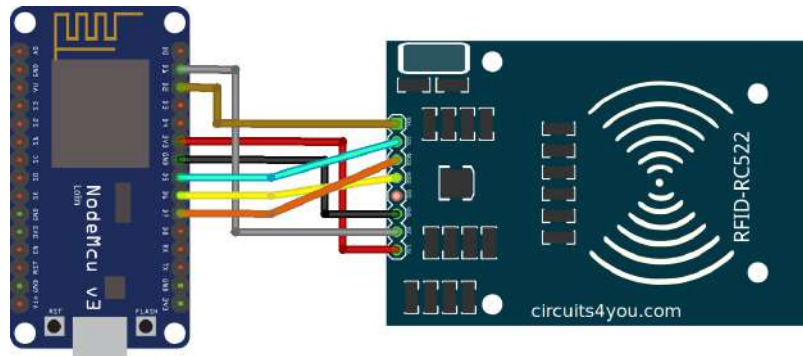


Figure 2.1 RFID

RFID belongs to a group of technologies referred to as Automatic Identification and Data Capture (AIDC). AIDC methods automatically identify objects, collect data about them, and enter those data directly into computer systems with little or no human intervention. RFID methods utilize radio waves to accomplish this. At a simple level, RFID systems consist of three components: an RFID tag or smart label, an RFID reader, and an antenna. RFID tags contain an integrated circuit and an antenna, which are used to transmit data to the RFID reader (also called an interrogator). The reader then converts the radio waves to a more usable form of data. Information collected from the tags is then transferred through a communications interface to a host computer system, where the data can be stored in a database and analyzed at a later time.

As stated above, an RFID tag consists of an integrated circuit and an antenna. The tag is also composed of a protective material that holds the pieces together and shields them from various environmental conditions. The protective material depends on the application. For example, employee ID badges containing RFID tags are typically made from durable plastic, and the tag is embedded between the layers of plastic. RFID tags come in a variety of shapes and sizes and are either passive or active. Passive tags are the most widely used, as they are smaller and less expensive to implement. Passive tags must be “powered up” by the RFID reader before they can transmit data. Unlike passive tags, active RFID tags have an onboard power supply (e.g., a battery), thereby enabling them to transmit data at all times. For a more detailed discussion, refer to this article: [Passive RFID Tags vs. Active RFID Tags](#).

Smart labels differ from RFID tags in that they incorporate both RFID and barcode technologies. They’re made of an adhesive label embedded with an RFID tag inlay, and they may also feature a barcode and/or other printed information. Smart labels can be encoded and printed on-demand using desktop label printers, whereas programming RFID tags are more time consuming and requires more advanced equipment.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Existing system

There are two ways of existing systems in the vehicle access of Mae Fah Luang University. The first way is the way to provide the car card. Giving the car card when entrance and getting back the car card when exit. The second way is students and employees have to register the car at the building division of Mae Fah Luang University. Students and employees who registered this access card will be able to enter and exit the university without having to exchange any card.

3.2 New System

3.2.1 Block Diagram

The Block Diagram identifies the system and component that was used in this study. The new system could be represented by a block diagram as shown in Figure 3.1

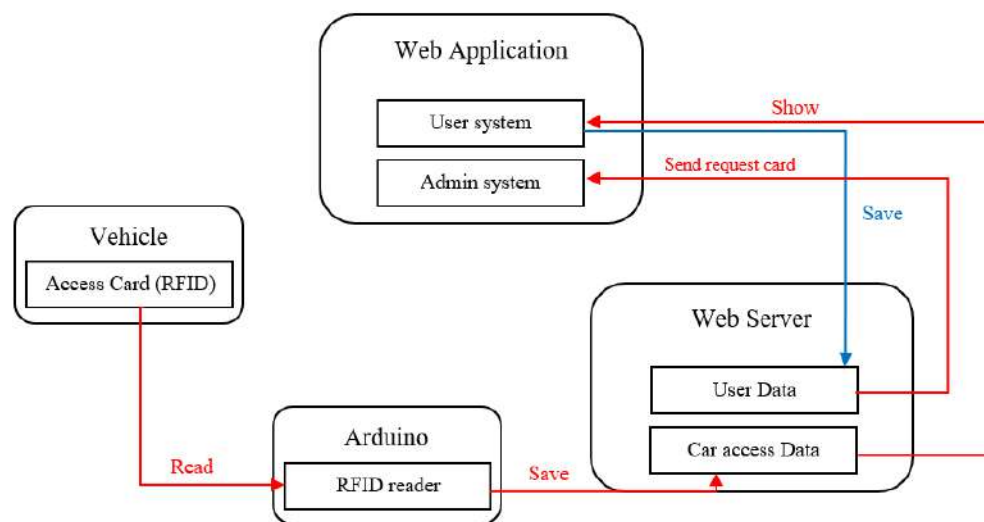


Figure 3.1 Block diagram of the system

From figure 3.1 this diagram shows all of the components in the project. Firstly, users have to register the account in a register form in a web application and then the data will be kept in the web server. The system will send the email to user for confirm the account. After confirming the user can login the web application. For the way to get the access card, user have to request an access card in user system. After requesting admin will check the information of user, provide the number of RFID and approve the request. User will get an access card from the MFU building division. For the part of entrance and exit, user have to scan their access card in to the RFID card reader and

then the data will be kept in the server. Finally, the data will be shown in the web application both of admin system and user system.

3.3 Activity diagram

The activity diagram identifies the system flow in each function. The new system could be represented by a activity diagram as shown in Figure 3.2-3.4

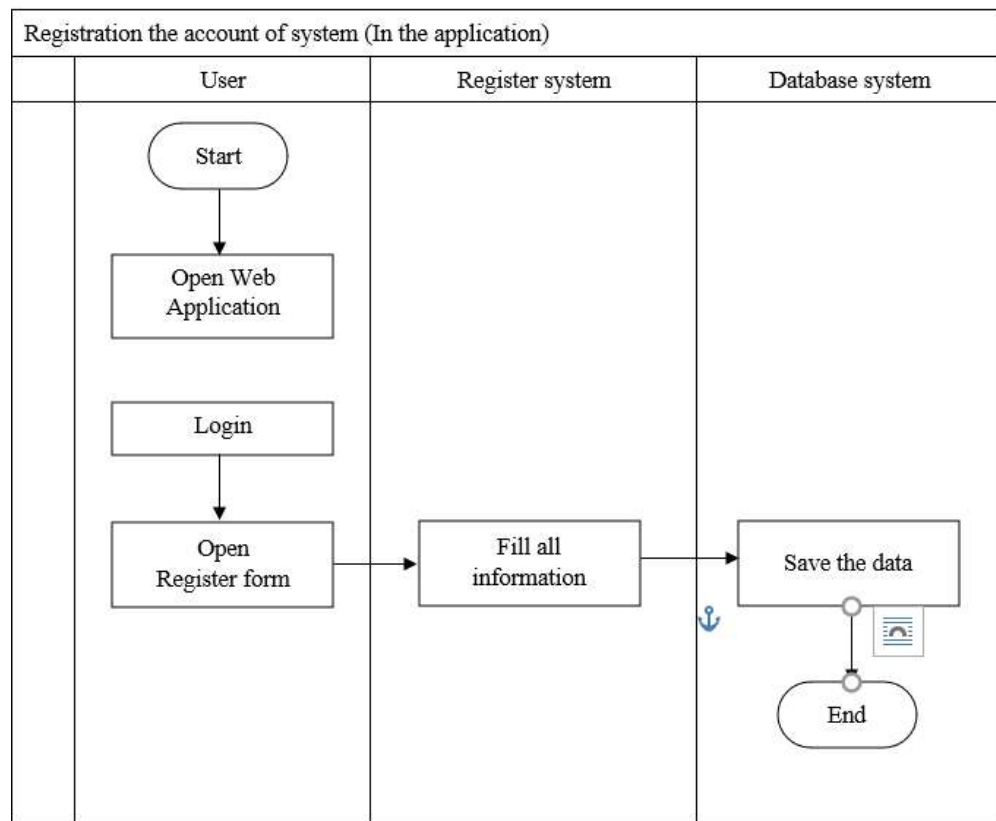


Figure 3.2 Activity diagram of Registration the account of web application

From figure 3.2 shows the activity diagram of the registration account of the web application. First, user opens the web application and then clicks to open the register system. The system will show the form that used to register. User have to fill in all the information that is required in the registration form and then click register. The system will send the link to confirm the account to user email

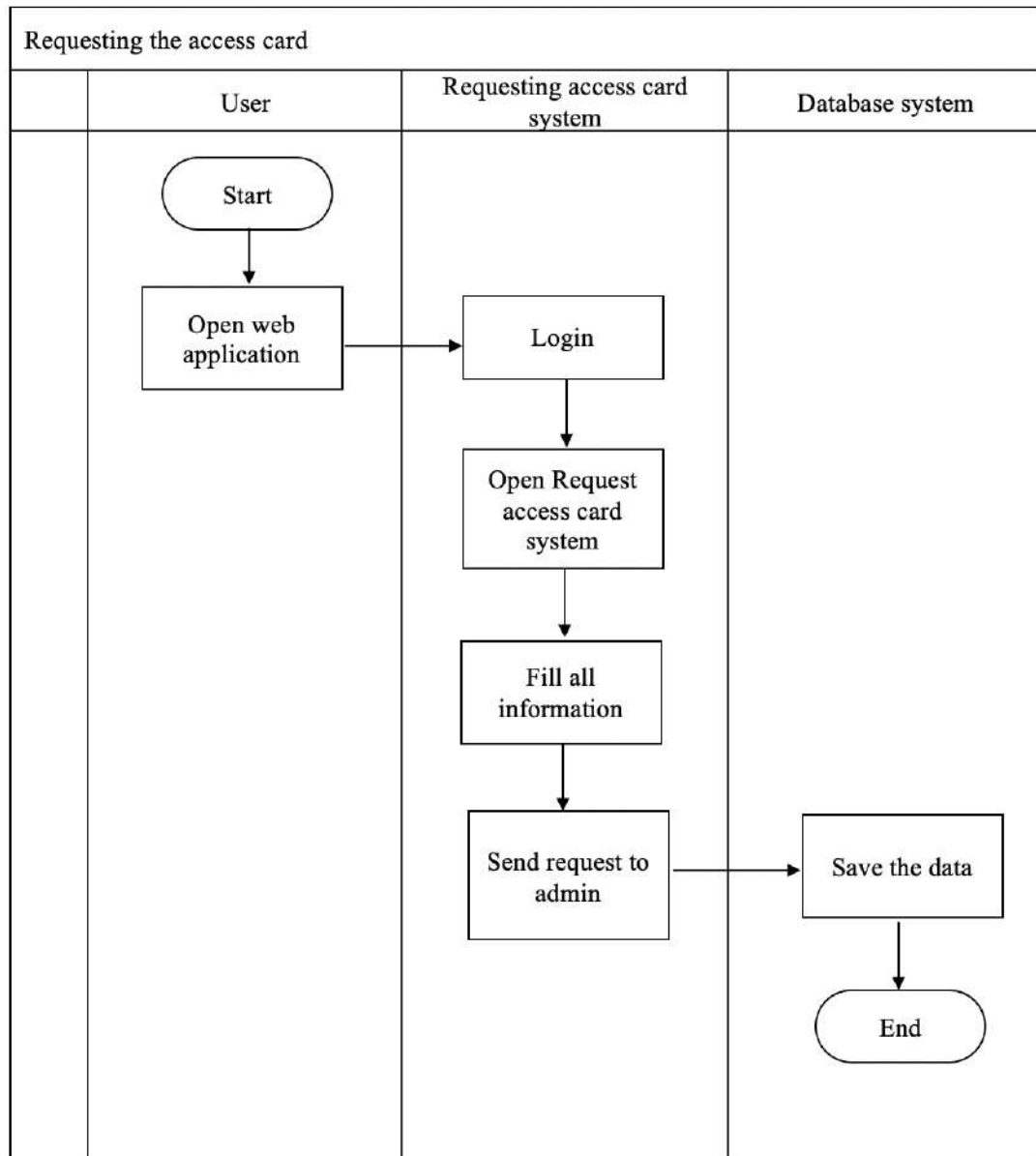


Figure 3.3 Activity diagram of requesting the access card

From figure 3.3 shows the activity diagram of requesting the access card. Firstly, users have to open a web application and login. Next, open the requesting access card form. Users have to fill in all the information that is required in the registration form and then click request. After that the data will be saved in the database waiting for approval.

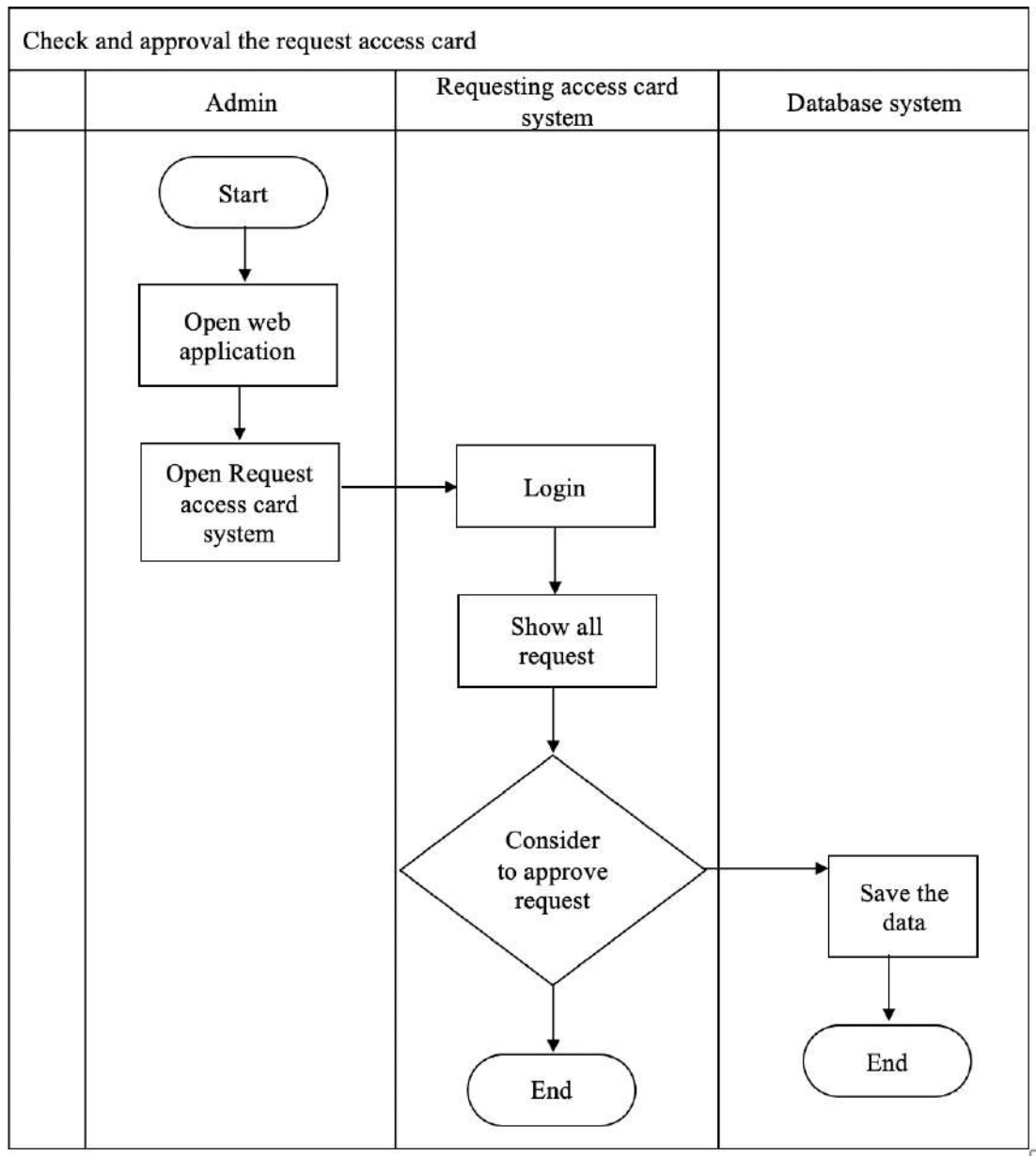


Figure 3.4 Activity diagram of check and approval the vehicle information

From figure 3.4 shows the activity diagram of check and approval of request access card. First, the admin will open a web application then open the admin system and login. The system will show all of the requested access. Admin will consider and check the information in the request. If all information is right, the admin will approve the request access and the data will save in the database. Admin will provide the data of RFID and send to next process.

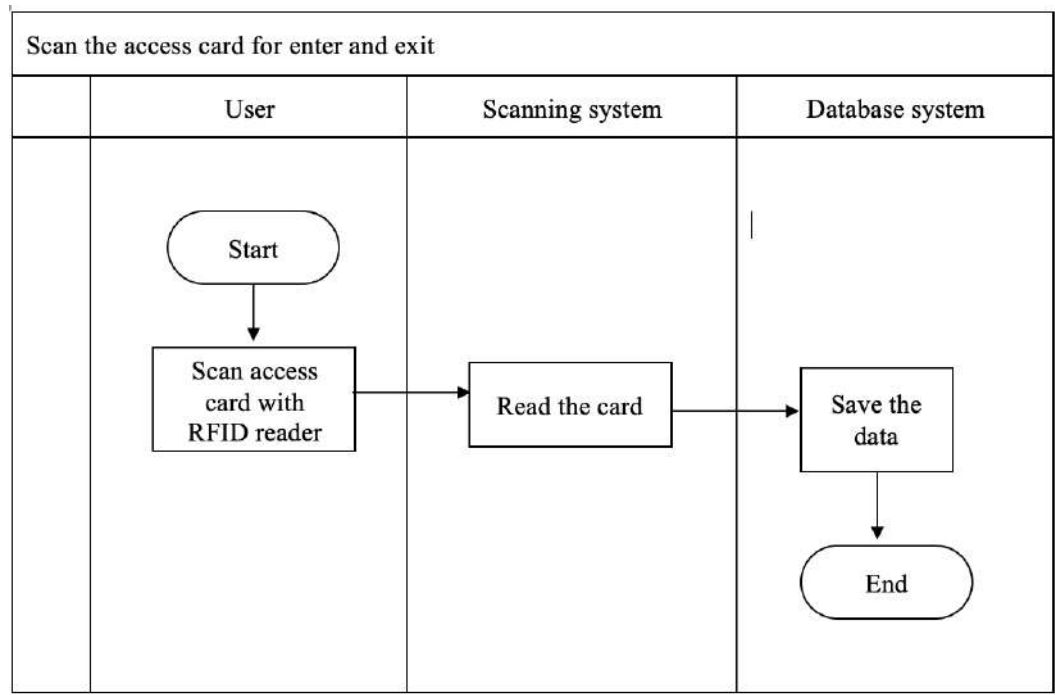


Figure 3.5 Activity diagram of scan access card

From figure 3.5 shows the activity diagram of the scan access card for enter and exit. First, the users scan access cards with RFID card readers. The sensor will read the card and then save the data in the database

3.4 Use case diagram

After gathering all of requirements, the new system could be represented by a use case diagram as shown in Figure 3.6

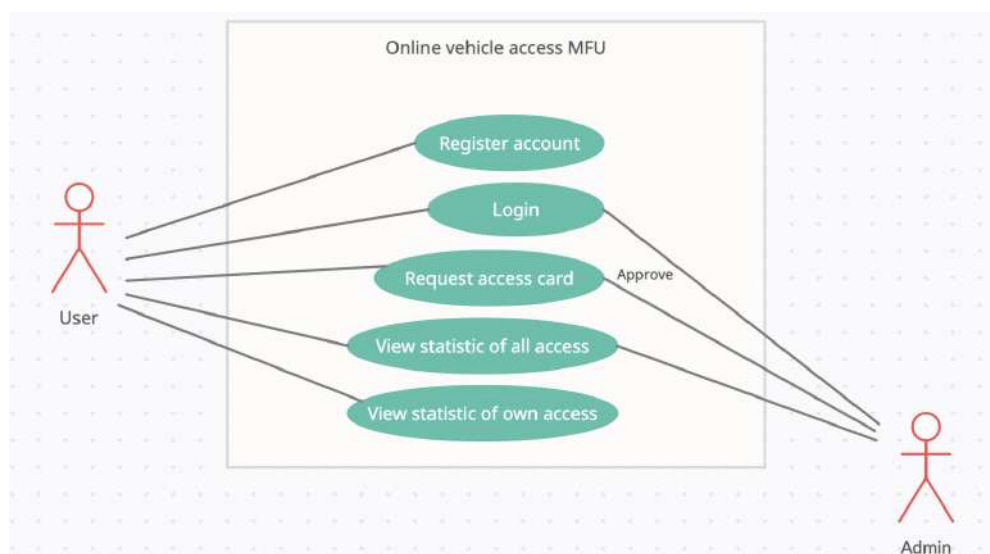


Figure 3.6 Use case diagram of online vehicle access monitoring of MFU

From figure 3.6 shows the use case diagram of online vehicle access monitoring of MFU users means vehicle owner. Users can register the account to the system and request the access card. Admin will view the request and approve the request and prepare the RFID for the user. And then user have to take the access card at MFU building division.

3.5 Database

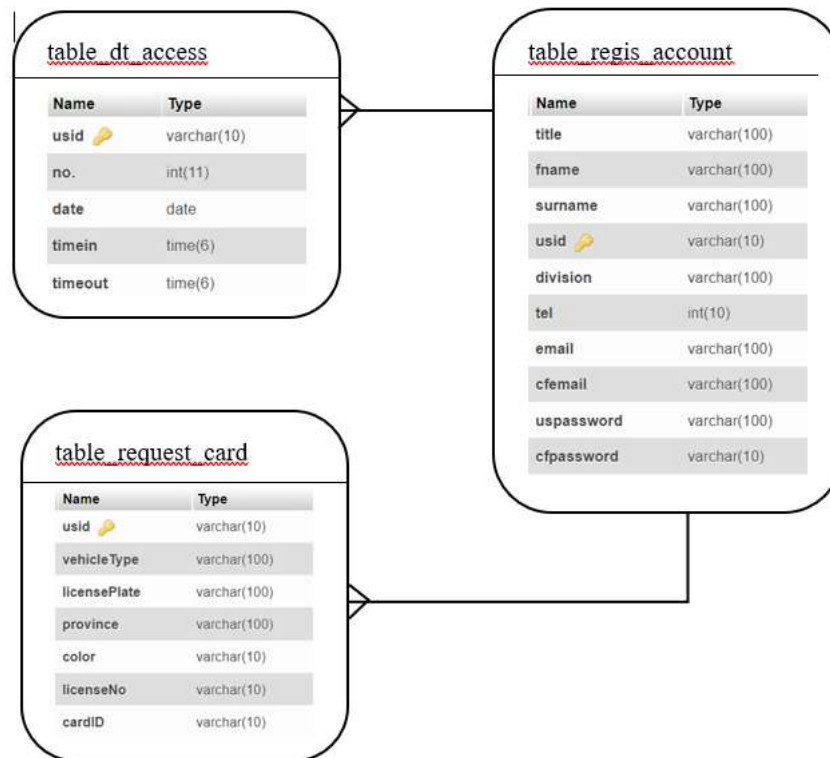
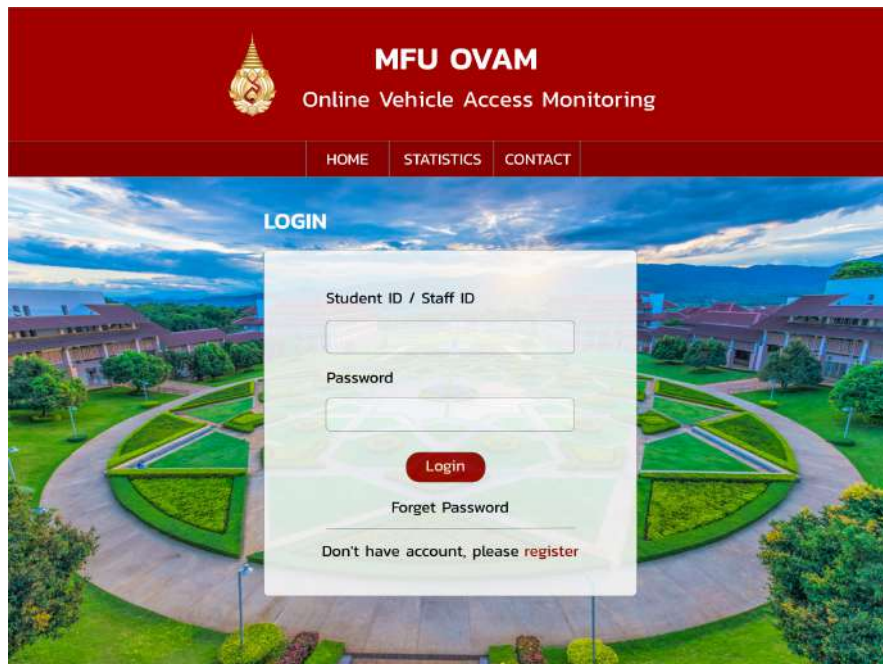


Figure 3.6 Database (ER-Diagram)

From figure 3.32 shows the structure of table in database and relationship of table **table_regis_account** keep the data of account for each user, **table_request_card** keep the data of request access card, **table_dt_access** keep the data of vehicle access.

3.6 Use Interface Design



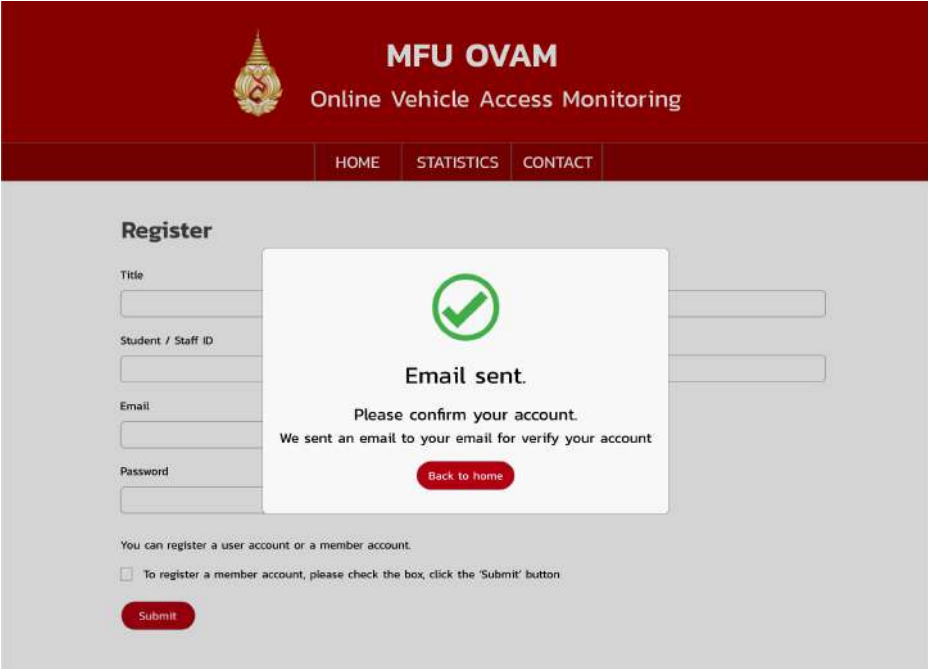
The screenshot shows the home page of the MFU OVAM (Online Vehicle Access Monitoring) system. The header is red with the MFU logo and the text "MFU OVAM Online Vehicle Access Monitoring". Below the header is a navigation bar with links for HOME, STATISTICS, and CONTACT. The main content area features a large background image of a school campus. Overlaid on this is a white login form with the title "LOGIN". The form contains two input fields: "Student ID / Staff ID" and "Password". Below these fields is a red "Login" button. Under the button are links for "Forget Password" and "Don't have account, please register".

Figure 3.7 Home page



The screenshot shows the registration form of the MFU OVAM system. The header is red with the MFU logo and the text "MFU OVAM Online Vehicle Access Monitoring". Below the header is a navigation bar with links for HOME, STATISTICS, and CONTACT. The main content area features a white registration form with the title "Register". The form contains several input fields: "Title" (a dropdown menu), "Name", "Surname", "Student / Staff ID", "School / Division", "Tel.", "Email", "Confirm Email", "Password", and "Confirm Password". Below the form is a checkbox labeled "To register a member account, please check the box, click the 'Submit' button". At the bottom of the form is a red "Submit" button.

Figure 3.8 Registration form



The image shows the MFU OVAM registration page. At the top, there is a red header with the MFU logo and the text "MFU OVAM Online Vehicle Access Monitoring". Below the header is a navigation bar with links for HOME, STATISTICS, and CONTACT. The main content area is titled "Register" and contains a registration form with fields for Title, Student / Staff ID, Email, and Password. A red "Submit" button is at the bottom of the form. A white pop-up box with a green checkmark icon and the text "Email sent. Please confirm your account. We sent an email to your email for verify your account" is centered over the form. A red "Back to home" button is at the bottom of the pop-up. Below the form, there is a note: "You can register a user account or a member account. To register a member account, please check the box, click the 'Submit' button".

Figure 3.9 Pop-up finished registration.



Figure 3.10 Page after login

MFU OVAM
Online Vehicle Access Monitoring

HOME STATISTICS CONTACT

Your Statistics
Your Information
Request access card
Log out

Search :

Sort :

Show : 7 per page << < 1 > >>

No.	Date	Time-in	Time-out	Note

Summary static

<< < 1 2 3 4 > >>

Figure 3.11 User access information

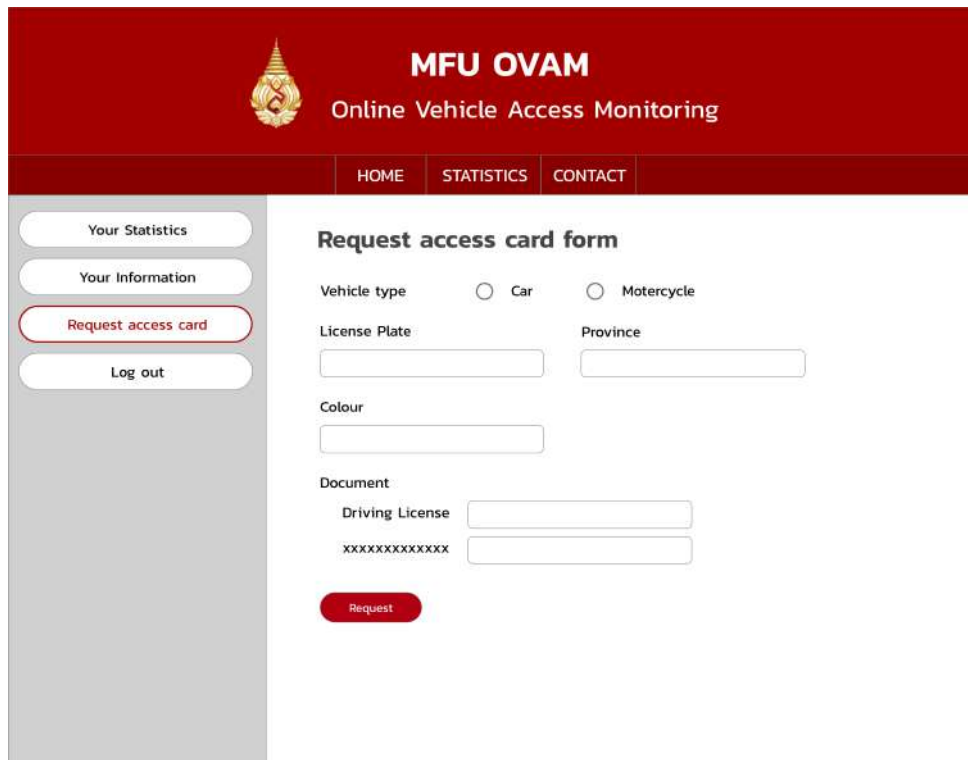
MFU OVAM
Online Vehicle Access Monitoring

HOME STATISTICS CONTACT

Your Statistics
Your Information
Request access card
Log out

User information
including approve status of the card

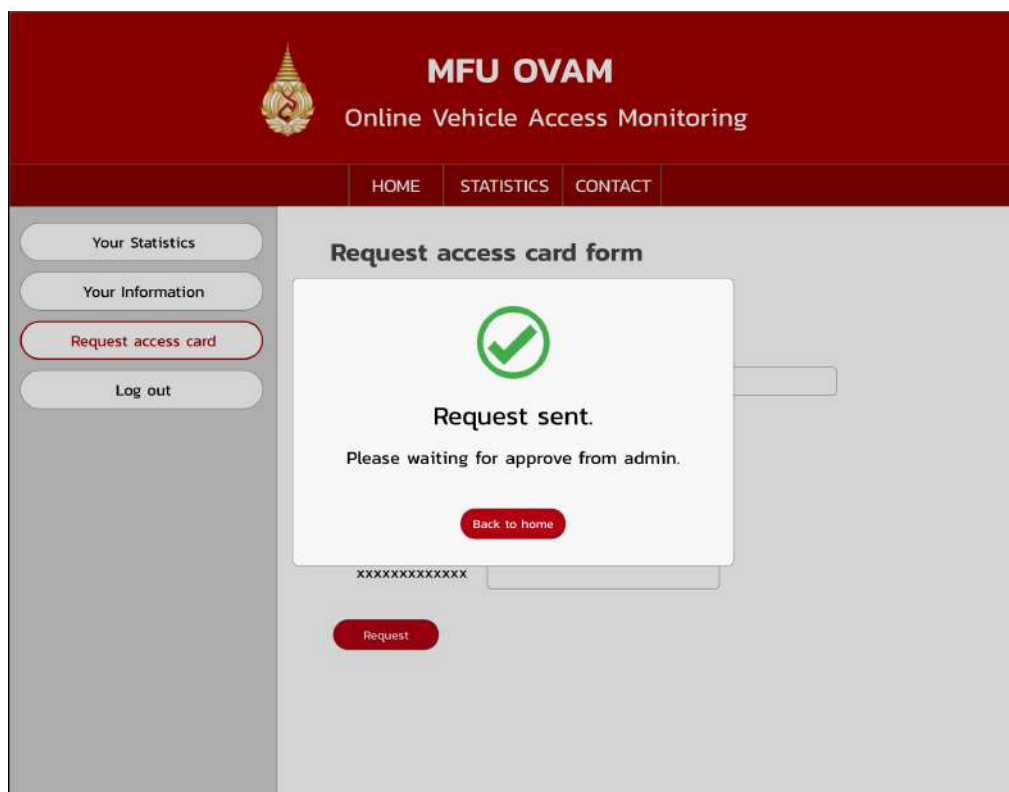
Figure 3.12 User information



The screenshot shows the MFU OVAM (Online Vehicle Access Monitoring) web application. The header is red with the MFU logo and the text 'MFU OVAM Online Vehicle Access Monitoring'. Below the header is a navigation bar with 'HOME', 'STATISTICS', and 'CONTACT' links. On the left side, there is a sidebar with four buttons: 'Your Statistics', 'Your Information', 'Request access card' (highlighted with a red border), and 'Log out'. The main content area is titled 'Request access card form' and contains the following fields and controls:

- Vehicle type:** Radio buttons for 'Car' and 'Motorcycle'.
- License Plate:** A text input field.
- Province:** A text input field.
- Colour:** A text input field.
- Document:** A section containing two text input fields. The first is labeled 'Driving License' and the second is labeled 'xxxxxxxxxxxx'.
- Request:** A red button at the bottom of the form.

Figure 3.13 Request access card form

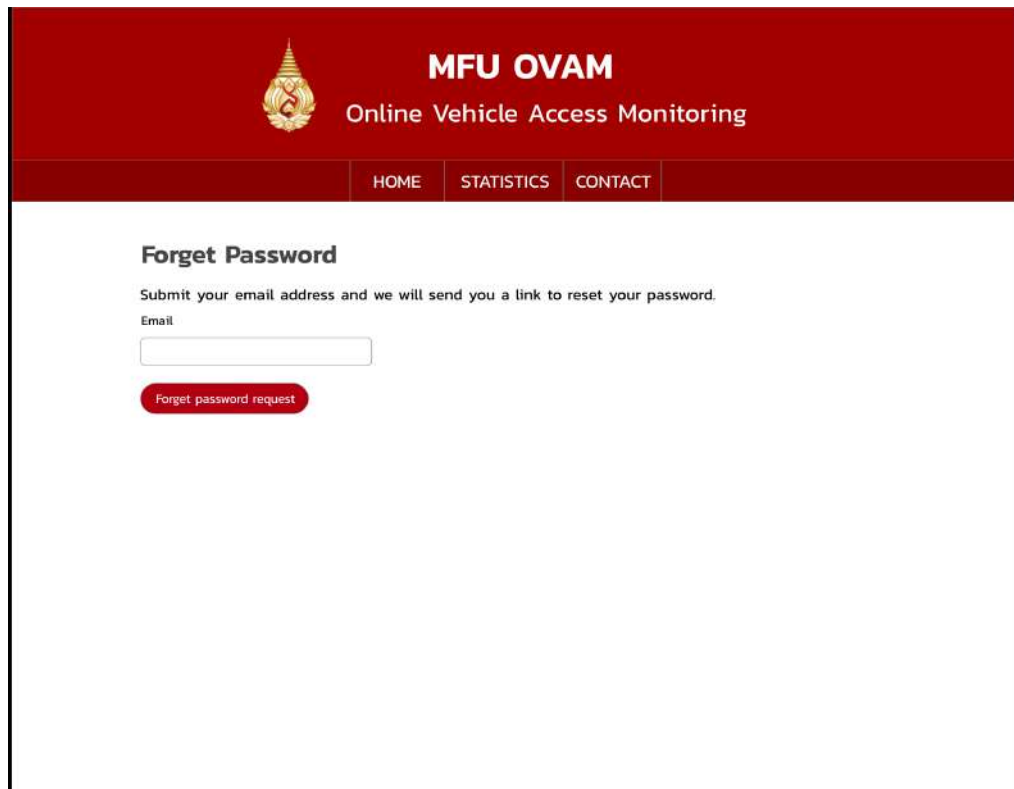


This screenshot shows the same MFU OVAM web application, but with a success pop-up displayed over the form. The pop-up is a white box with a green checkmark icon and the following text:

- Request sent.**
- Please waiting for approve from admin.
- Back to home:** A red button.

The background form is dimmed. The sidebar and navigation bar remain the same as in Figure 3.13.

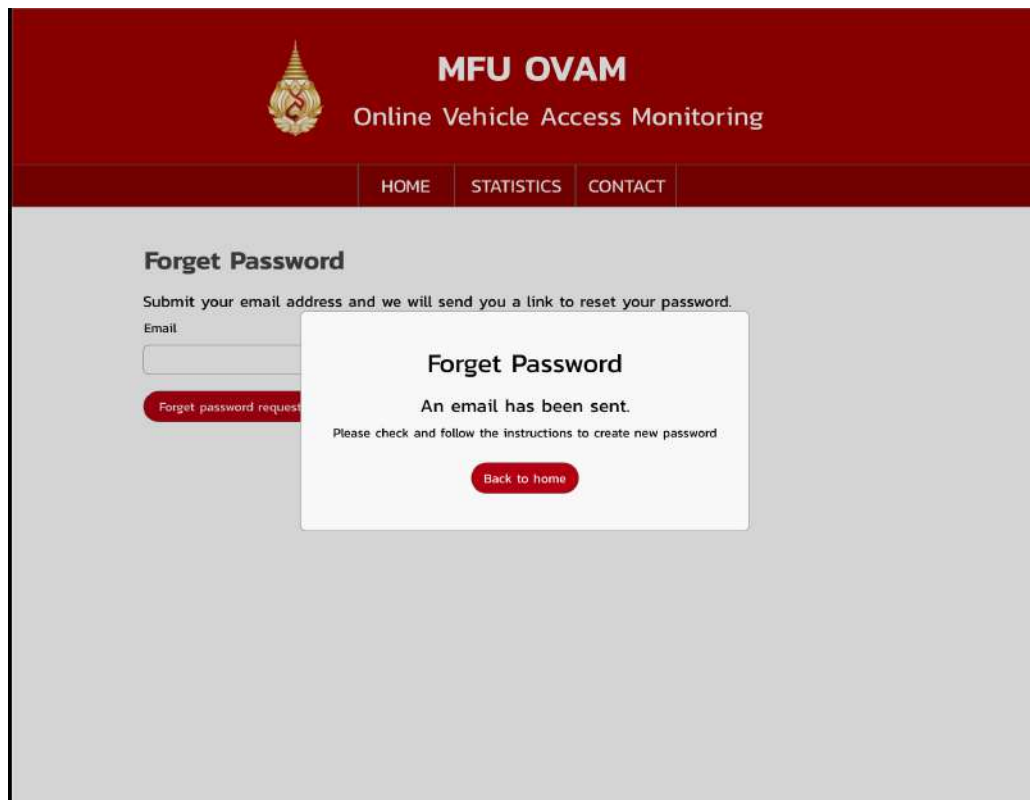
Figure 3.14 Pop-up Request access card form



The screenshot shows the 'Forget Password' form on the MFU OVAM website. The header is dark red with the MFU logo and the text 'MFU OVAM Online Vehicle Access Monitoring'. Below the header is a navigation bar with 'HOME', 'STATISTICS', and 'CONTACT' links. The main content area is white and contains the following elements:

- Forget Password** (Section Header)
- Submit your email address and we will send you a link to reset your password.
- Email (Label)
- (Email input field)
- [Forget password request](#) (Red button)

Figure 3.15 Forget password

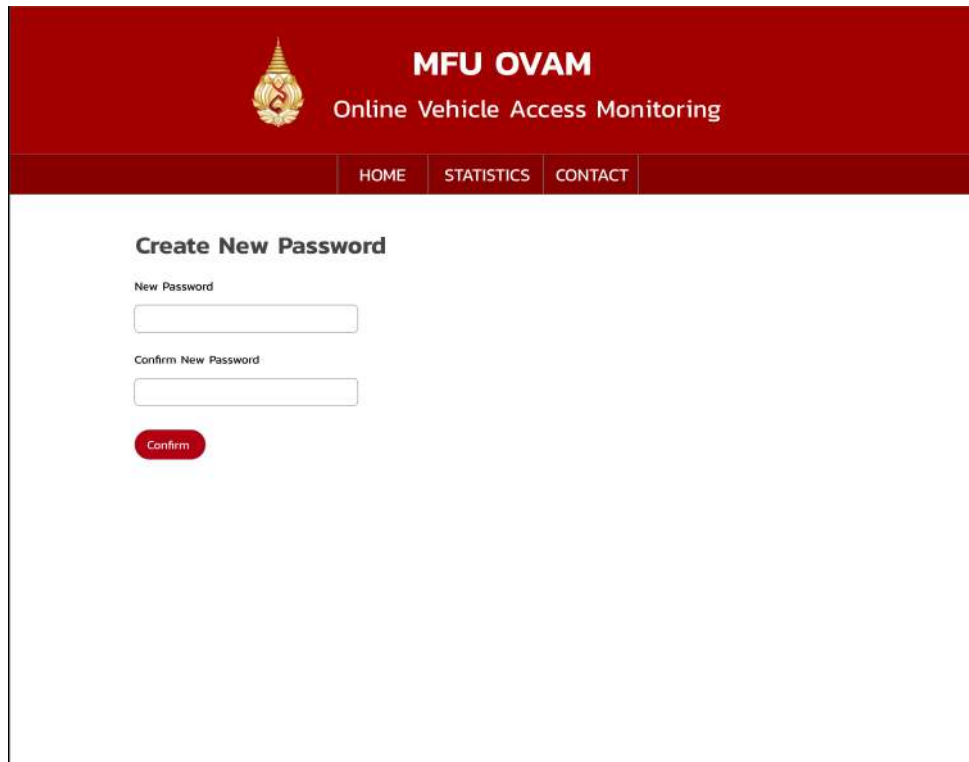


This screenshot shows the same 'Forget Password' form as Figure 3.15, but with a success message pop-up displayed. The pop-up is a white box with a red border and contains the following text:

- Forget Password** (Section Header)
- An email has been sent.
- Please check and follow the instructions to create new password
- [Back to home](#) (Red button)

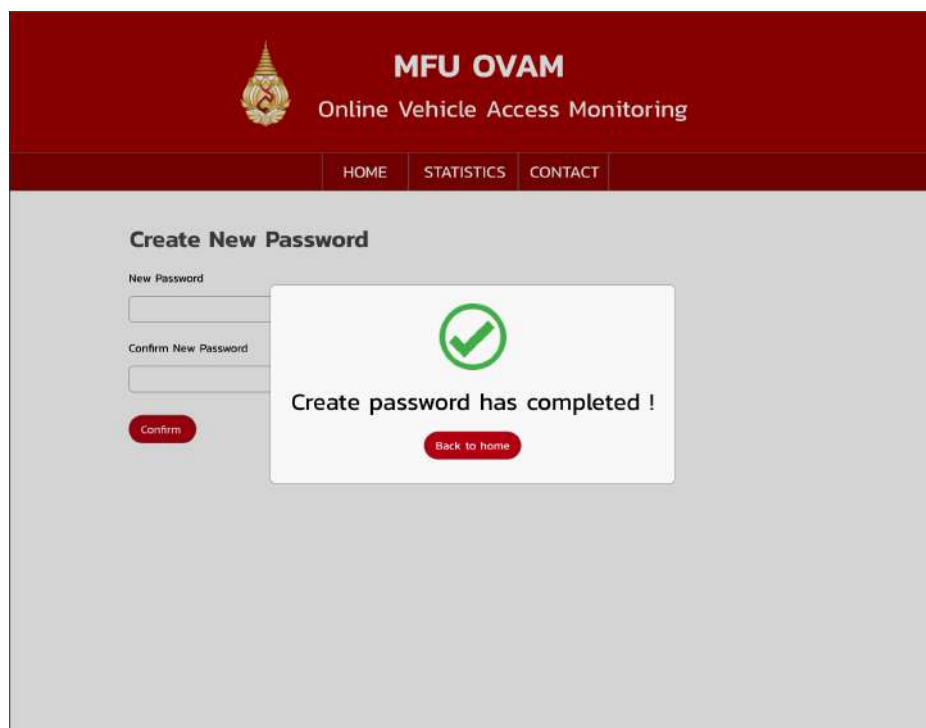
The background of the form is now a light gray, and the 'Email' input field and 'Forget password request' button are partially obscured by the pop-up.

Figure 3.16 Pop-up Forget password



The screenshot shows the 'Create New Password' form on the MFU OVAM website. The header is dark red with the MFU logo and the text 'MFU OVAM Online Vehicle Access Monitoring'. Below the header is a navigation bar with 'HOME', 'STATISTICS', and 'CONTACT' links. The form itself is white and contains two input fields: 'New Password' and 'Confirm New Password'. A red 'Confirm' button is located below the second field.

Figure 3.17 Create new password



This screenshot shows the same 'Create New Password' form as Figure 3.17, but with a success message pop-up. The pop-up is a white box with a green checkmark icon and the text 'Create password has completed !'. Below the text is a red 'Back to home' button. The background of the form is now light gray.

Figure 3.18 Pop-up Create new password

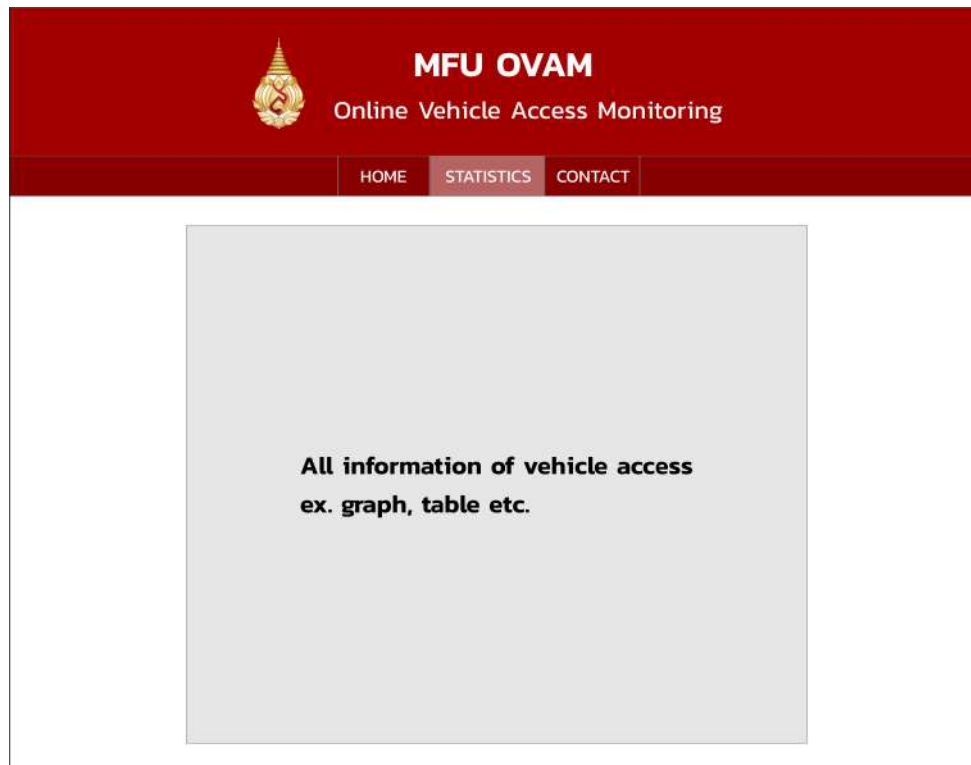


Figure 3.19 Statistic of all vehicle access

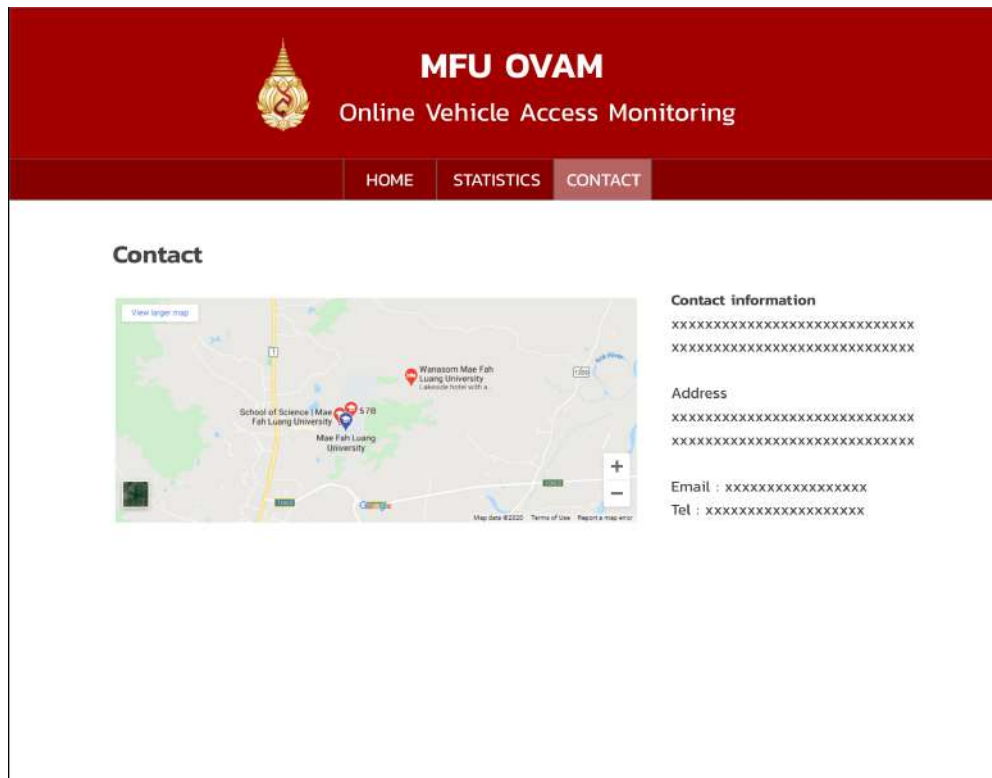


Figure 3.20 Contact

CHAPTER 4

EXPERTIMENT AND RESULT

4.1 Hardware

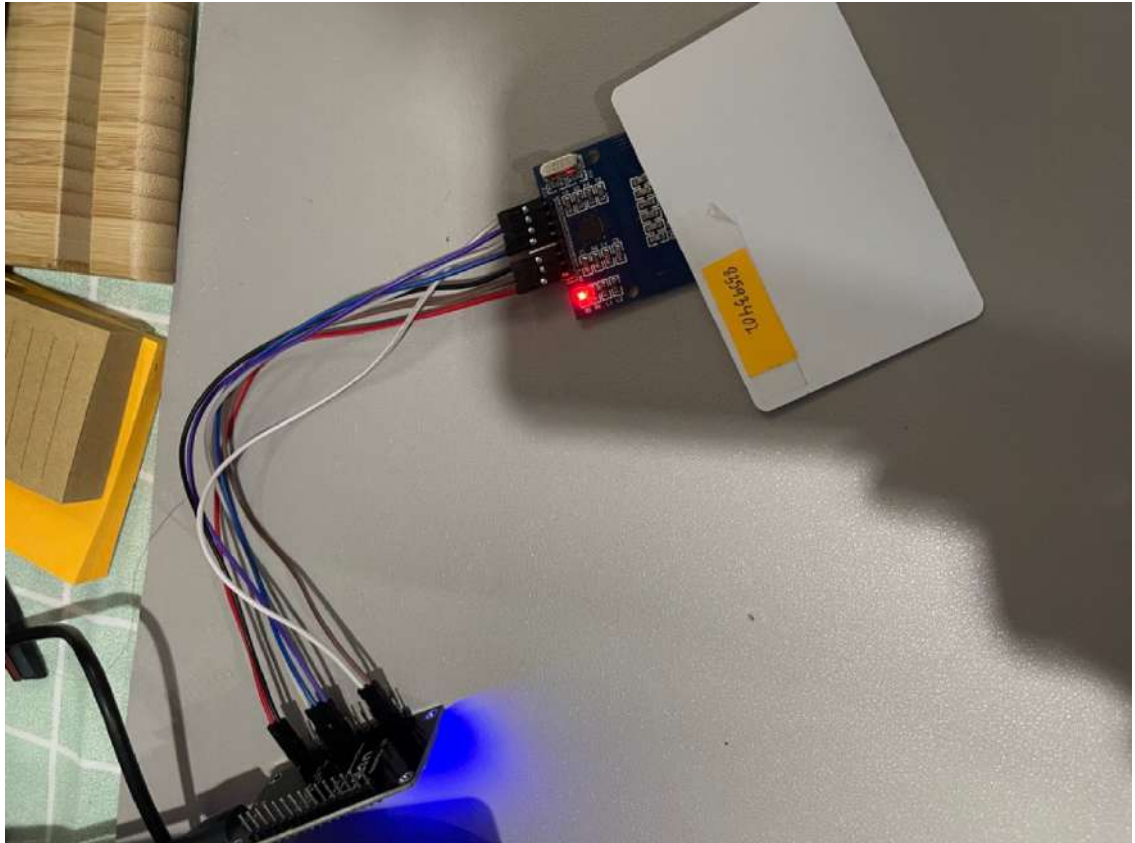
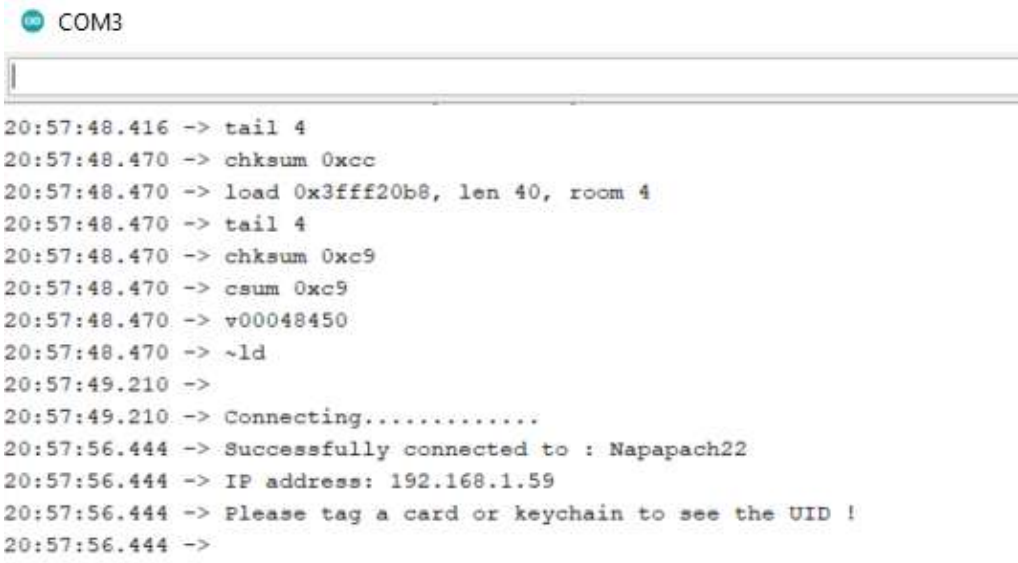


Figure 3.21 NodeMCU ESP8266 + RC522

From figure 3.21 shows NodeMCU ESP 8266 + RC522 connect port is showing below:

D2	=	SDA/SS
D5	=	SCK
D7	=	MOSI
D6	=	MISO
GND	=	GND
D1	=	RST
3V/3V3	=	3.3V



```

COM3

20:57:48.416 -> tail 4
20:57:48.470 -> chksum 0xcc
20:57:48.470 -> load 0x3fff20b8, len 40, room 4
20:57:48.470 -> tail 4
20:57:48.470 -> chksum 0xc9
20:57:48.470 -> csum 0xc9
20:57:48.470 -> v00048450
20:57:48.470 -> ~ld
20:57:49.210 ->
20:57:49.210 -> Connecting.....
20:57:56.444 -> Successfully connected to : Napapach22
20:57:56.444 -> IP address: 192.168.1.59
20:57:56.444 -> Please tag a card or keychain to see the UID !
20:57:56.444 ->

```

Figure 3.33 Arduino IDE Serial monitor

From figure 3.33 shows the Serial Monitor of Arduino IDE. The Arduino IDE connected with NodeMCU ESP8266 that connected with RFID-RC522. The board use for read the card id of access card. Serial monitor shows date/time that the card was stamp and the card id number.

4.2 User Web Application

Web application will separate to two part including user part and admin part. And both of web application can support many devices for example computer, mobile phone and tablet.

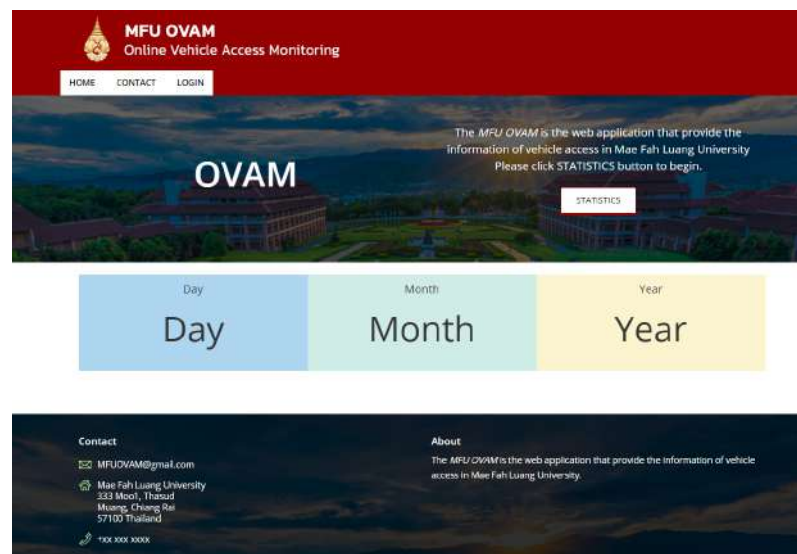


Figure 3.21 Home Page (User part)

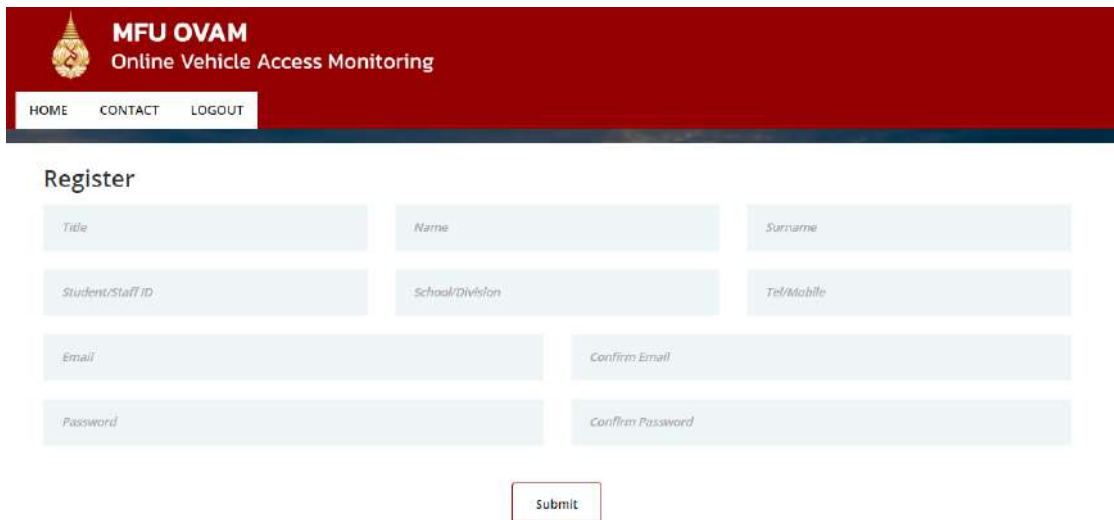
From figure 3.21 shows the first page of web application. The home page provides the number of car access in three format Day, Month, Year. And user can access other page by using navigator menu in the top of page. In the footer of page provide short information of contact admin.

Figure 3.22 Contact Page (User part)

From figure 3.22 shows the form of contact user and the information of contact admin including in text and google map will provide the location.

Figure 3.23 Login Page (User part)

From figure 3.23 shows the form of login. User have to login in this page before go to the page that provide user information. If the user does not have account, user have to register the account by click 'Register' link.



MFU OVAM
Online Vehicle Access Monitoring

HOME CONTACT LOGOUT

Register

Title Name Surname

Student/Staff ID School/Division Tel/Mobile

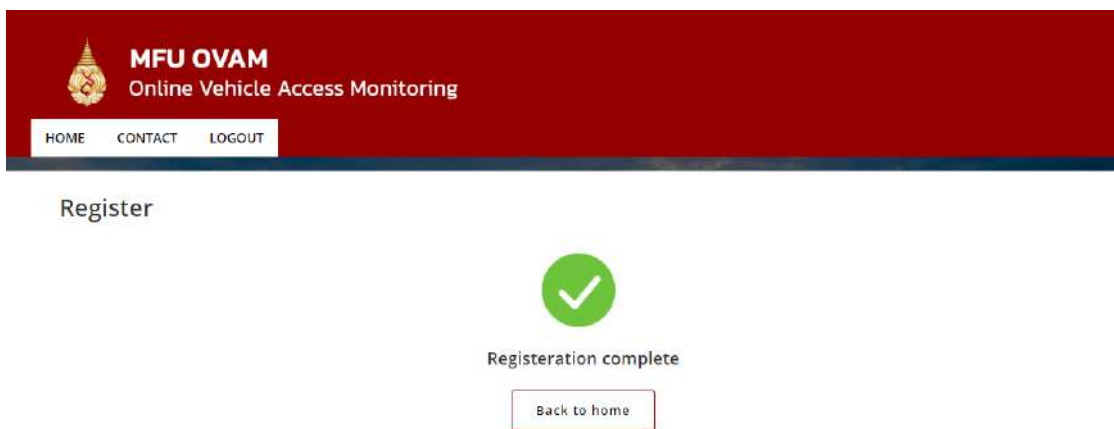
Email Confirm Email

Password Confirm Password

Submit

Figure 3.24 Register Page (User part)

From figure 3.24 show the form of register. User have to register the account in this page before request the access card. After user register the account, the information that user fills in the form will keep inn the database. And then user can login the web application to see their own information.



MFU OVAM
Online Vehicle Access Monitoring

HOME CONTACT LOGOUT

Register

Registration complete

Back to home

Figure 3.25 Completed register page (User part)

From figure 3.25 show the completed of register. This page has action to insert the information that user fills in the form save into the database. If user wants to login, user can click logout or back to home button.

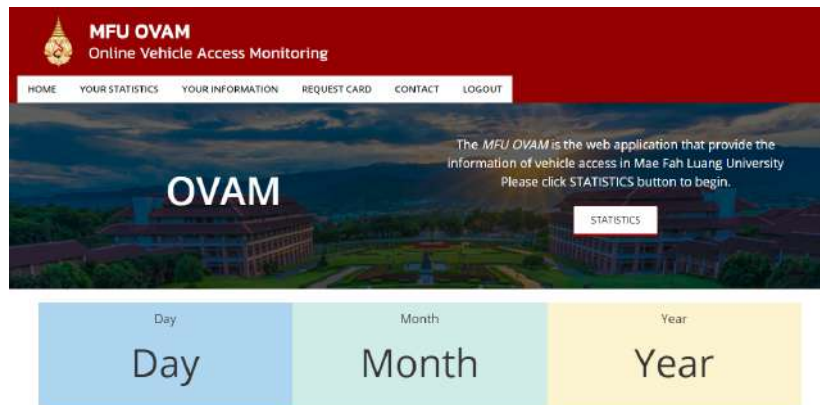


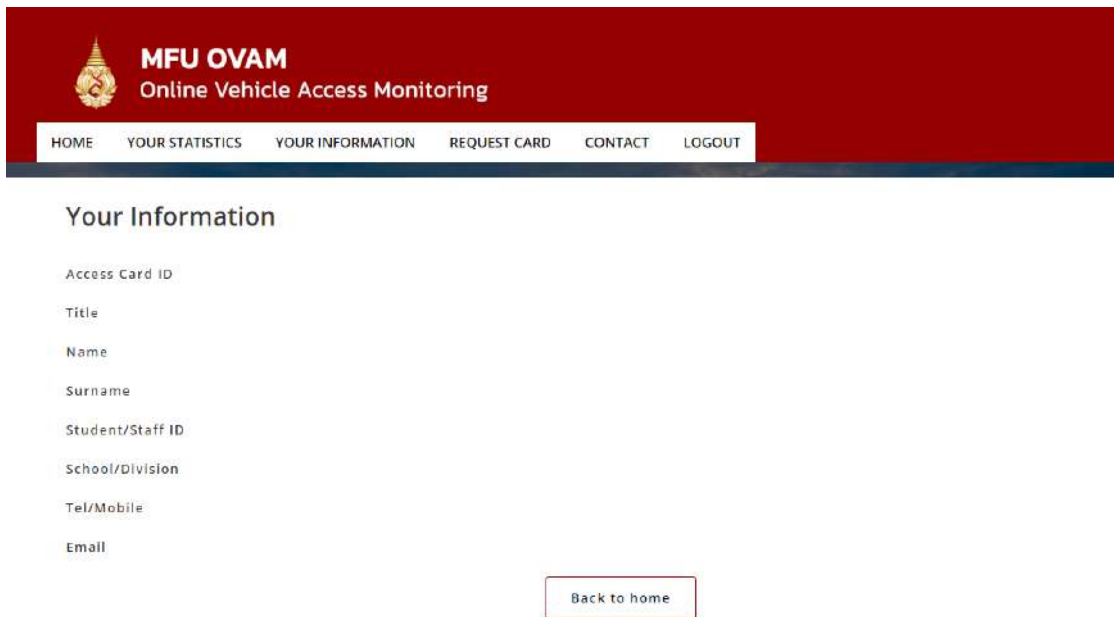
Figure 3.26 Home page after login (User part)

From figure 3.26 shows the home page after login. The page provides the number of car access like the home page before login but this page changes the navigation menu to have the menu that user can access own information.

NO.	DATE	TIME-IN	TIME-OUT
1	30-05-2021	17.43.00	18.00.00

Figure 3.27 Your statistics page (User part)

From figure 3.27 show the in statistic of access Mae Fah Luang University of user. The page provides date and time of access.



MFU OVAM
Online Vehicle Access Monitoring

HOME YOUR STATISTICS YOUR INFORMATION REQUEST CARD CONTACT LOGOUT

Your Information

Access Card ID

Title

Name

Surname

Student/Staff ID

School/Division

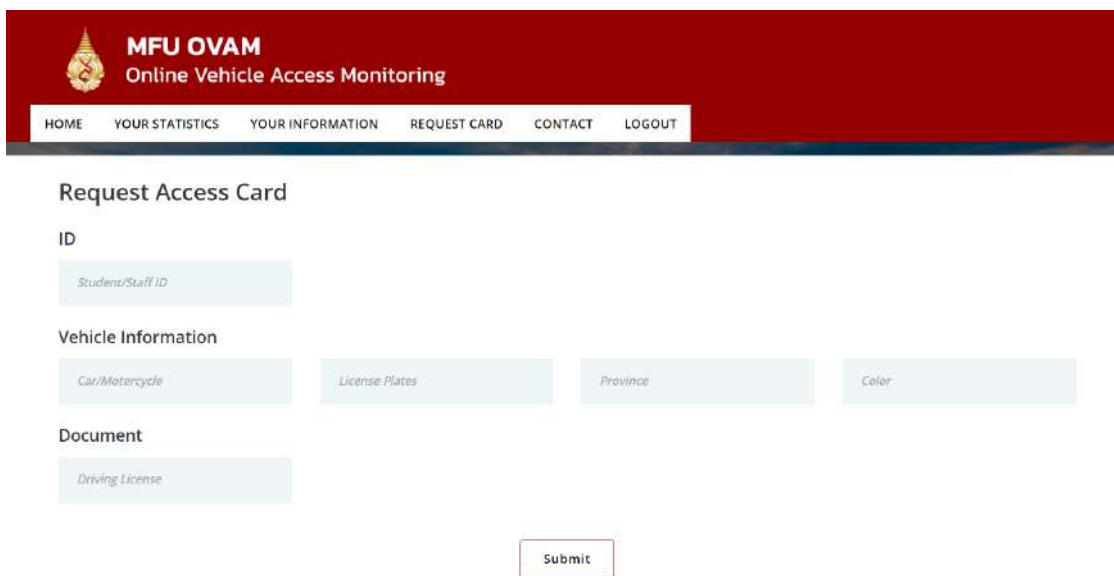
Tel/Mobile

Email

Back to home

Figure 3.28 Your information (User part)

From figure 3.28 show the information of user that user fill in the form in register page. If user already get the access card the page will provide information of Access Card ID.



MFU OVAM
Online Vehicle Access Monitoring

HOME YOUR STATISTICS YOUR INFORMATION REQUEST CARD CONTACT LOGOUT

Request Access Card

ID

Student/Staff ID

Vehicle Information

Car/Motorcycle License Plates Province Color

Document

Driving License

Submit

Figure 3.29 Request access card page (User part)

From figure 3.29 shows the form of request access card. Before get the access card, user have to fill the request access card form. When user click submit button, the data will save in the database and send the data to Admin page. Admin will provide the

id of RFID and fill in in data of id then the data of id will show in your information page.

4.3 Admin Web Application

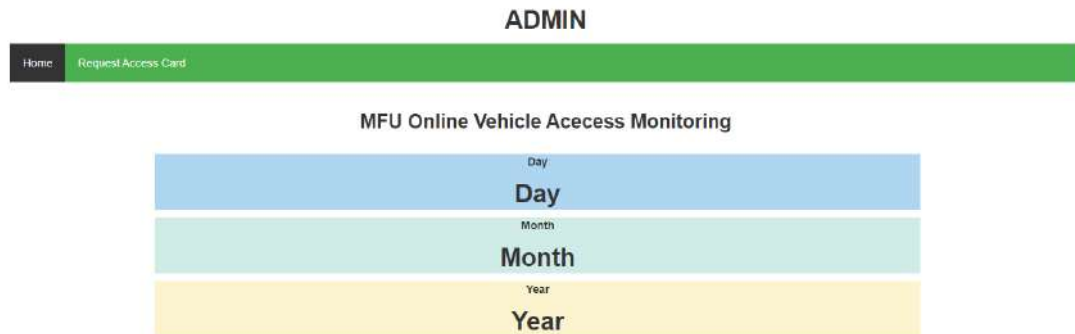


Figure 3.30 Home page (Admin part)

From figure 3.30 shows the home page of admin. The page provides the number of car access like the home page of user but this page changes the navigation menu to have the menu that admin manage request of access card



Figure 3.31 Request Access Card page (Admin part)

From figure 3.31 shows the list of request access card. Admin have to edit row to add Access card ID. And then, the data of access card id will keep in the database and show in your information page of user.

CHAPTER 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

In this project, the system uses the equipment (NodeMCU ESP8266) connected with the sensor that can help admin to read and keep data of vehicle access easily. Moreover, the system will reduce the traffic jam around the entrance and exit of Mae Fah Luang University. Because user just stamp the card with the board and then the data will auto to save in the database. After that the user can see the vehicle access data in summary and their own data in the web application.

For future work, the system will update to keep more field of data. And change the equipment to Bluetooth that help user to stamp the card farther that RFID.

REFERENCES**REFERENCES**

[1] Reangha, K . , & Yangngam, N . , & Rangsiyopas, V .(2012). RMUTT Vehicle license plates detection and information system.

[2] ชัยวัช มินเดินเรือ.(2016). Automatic License Plate Recognition System .

[3] Smart campus Management center of CMU(2018).บัตรอนุญาตผ่านเข้า-ออกมหาวิทยาลัยเชียงใหม่.
Retrieved 30 March 2021 from <https://scmc.cmu.ac.th/register>